THE ALGEBRAIC CRITERION FOR BEING PRO-ISOMORPHIC TO A DELIGNE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4S (sources/stacks-duality/pinned-original-source.tex line 8320).

Allowed standard prerequisites (editorial).

- The Arin-Rees consequence identifying the pro-systems of cohomology. Same chapter, Lemma labelled lemma-consequence-Artin-Rees-bis, cited at proof lines 8343-8345. Gives the forward direction.
- Truncation triangles. Derived Categories, Remark labelled derived-remark-truncation-distinguished-triangle, cited at proof lines 8349-8353. Supplies the compatible system of triangles for the induction.
- The 2-out-of-3 property for Deligne systems. Same chapter, Lemma labelled lemma-deligne-system-2-out-of-3, cited at proof lines 8353-8355. Concludes the induction step.

Source reference. Original lemma 0G4R: sources/stacks-duality/pinned-original-source.tex, source lines 8284–8284. Same-chapter pointer into the author environment lemma at source line 8284; invoked at proof line 8340 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8351 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4Q: sources/stacks-duality/pinned-original-source.tex, source lines 8236–8236. Same-chapter pointer into the author environment lemma at source line 8236; invoked at proof line 8354 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes. Let $a \leq b$ be integers. Let (K_n) be an inverse system of $D^b_{Coh}(\mathcal{O}_X)$ such that $H^i(K_n) = 0$ for $i \notin [a, b]$. The following are equivalent*

1. *(K_n) is pro-isomorphic to a Deligne system,*
2. *for every $p \in \mathbf{Z}$ there exists a coherent $\mathcal{O}_X$-module $\mathcal{F}$ such that the pro-systems $(H^p(K_n))$ and $(\mathcal{I}^n\mathcal{F})$ are pro-isomorphic.*

Proof. Assume (1). To prove (2) holds we may assume (K_n) is a Deligne system. By definition we may choose a bounded complex $\mathcal{F}^\bullet$ of coherent $\mathcal{O}_X$-modules and a quasi-coherent sheaf of ideals cutting out $X \setminus U$ such that K_n is represented by $\mathcal{I}^n\mathcal{F}^\bullet$. Thus the result follows from Lemma 0G4R.

Assume (2). We will prove that (K_n) is as in (1) by induction on $b - a$. If $a = b$ then (1) holds essentially by assumption. If $a < b$ then we consider the compatible system of distinguished triangles

$$\tau_{\leq a}K_n \to K_n \to \tau_{\geq a+1}K_n \to (\tau_{\leq a}K_n)[1]$$

See Derived Categories, Remark 08J5. By induction on $b - a$ we know that $\tau_{\leq a}K_n$ and $\tau_{\geq a+1}K_n$ are pro-isomorphic to Deligne systems. We conclude by Lemma 0G4Q. □